

推算每個月的開始是星期幾(佔分 20 分)

問題描述：

給定一組 12 個隨機排列的整數，這些數字代表某一年中每個月的起始星期。你的任務是根據這些數字和公曆月份天數，合理地推算每個數字對應的月份，並按照年曆順序（1 月至 12 月）輸出每個月的起始星期。

注意事項：

1. 閏年的 2 月有 29 天，平年則是 28 天。
2. 閏年為合乎下列規則之一的年份：
 - a. 閏年除以 4 能整除，且除以 100 不能整除。
 - b. 閏年剛好是 100 的倍數，且除以 400 能整除。
3. 需判別該數組是否能構成有效的年曆排列。
4. 1 代表星期一，2 代表星期二、...、7 代表星期日。

輸入說明：

輸入一行包含 12 個整數，數字之間用空格分隔，這些數字隨機排列，代表某一年中每個月的開始是星期幾。

輸出說明：

1. 若有合理解答：輸出一行包含 12 個整數，數字之間用空格分隔，依次表示從 1 月到 12 月，每個月的第一天是星期幾。
2. 若無合理解答：輸出 "Impossible" 。

範例 1：

輸入：

3 2 6 4 6 6 2 3 5 7 1 1

輸出：

3 6 6 2 4 7 2 5 1 3 6 1

範例 2：

輸入：

7 4 7 1 3 5 2 7 5 6 2 6

輸出：

Impossible

Deduce the Starting Day of the Week for Each Month (20 points)

Problem Description:

Given a set of 12 randomly arranged integers, these numbers represent the starting day of the week for each month in a certain year. Your task is to reasonably deduce the corresponding month for each number based on these numbers and the number of days in the Gregorian calendar months, and output the starting day of the week for each month in chronological order (January to December).

Notes:

1. February has 29 days in a leap year and 28 days in a common year.
2. A leap year is a year that meets one of the following rules:
 - a. The year is divisible by 4 but not divisible by 100.
 - b. The year is a multiple of 100 and is divisible by 400.
3. It is necessary to determine whether the array can form a valid calendar arrangement.
4. 1 represents Monday, 2 represents Tuesday, ..., 7 represents Sunday.

Input Description:

Input a line containing 12 integers separated by spaces. These numbers are randomly arranged and represent the starting day of the week for each month in a certain year.

Output Description:

1. If there is a reasonable solution: output a line containing 12 integers separated by spaces, sequentially representing the starting day of the week for the first day of each month from January to December.
2. If there is no reasonable solution: output "Impossible".

Example 1:

Input:

3 2 6 4 6 6 2 3 5 7 1 1

Output:

3 6 6 2 4 7 2 5 1 3 6 1

Example 2:

Input:

7 4 7 1 3 5 2 7 5 6 2 6

Output:

Impossible